Role-Play Driven UX: A TTRPG-Inspired Toolkit for User Needs Analysis

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Background

Tabletop role-playing games (TTRPGs) are structured yet highly imaginative play experiences in which participants take on fictional roles, make consequential decisions, and collaboratively explore scenarios. A typical TTRPG session combines formal rules with open-ended storytelling, allowing players to negotiate outcomes together rather than simply follow a fixed script. It creates space for participants to express goals, values, and emotions in narrative contexts.

Building on Michael Heron et al.'s work, we understand that these qualities make TTRPGs a promising method for user experience research and requirements elicitation, not just a form of entertainment. In their article, Heron and colleagues introduce the idea of "playable UX worlds" and a Requirements Elicitation Campaign Setting (RECS), where personas are treated as player characters, user journeys are framed as adventures, and character sheets, abilities, and encounters are tailored to the digital product under study. By positioning researchers as game masters and inviting participants to "play through" UX scenarios, their approach shows how TTRPG structures can surface more authentic needs, breakdowns, and emotional responses than conventional interviews, questionnaires, or scripted usability tasks often do.

Inspired by this perspective, our project explores how a lightweight, TTRPG-inspired toolkit can be brought into early-stage UX work. Specifically, we aim to translate key TTRPG concepts, such as character creation, attribute checks, and narrative encounters into Figma-based materials that can be used in user needs analysis and usability testing. In doing so, we hope to examine whether this gamified, narrative-driven approach can make the research process more engaging for participants and help researchers uncover richer, more situated insights about user behavior and motivations.

Objectives

The overall goal of this project is to investigate how TTRPG-inspired gamification can enhance traditional user needs analysis and early UX research activities.

Firstly， we will design a TTRPG-inspired UX research toolkit by creating a simple "player sheet" that turns personas and user skills into TTRPG-style characters, adding dice-based challenges linked to important steps in using the app, and writing short story-like scenarios that turn usability tasks into small narrative encounters.

Then we will use this toolkit in common UX activities, such as usability testing, user interviews, and persona creation, and observe how it affects participants' engagement, enjoyment, and the way they reflect on their experiences.

Finally, we will compare the insights gained from these TTRPG-style sessions with those from more traditional methods, and from this comparison, we aim to develop practical guidelines on when and how TTRPG-inspired approaches are most useful in UX research.

Methodology

Our study is guided by three main research questions.

First, we ask how a TTRPG-inspired UX toolkit changes the depth and richness of user needs we can find, compared with more traditional user research methods (RQ1).

Second, we want to know how participants experience these TTRPG-style sessions in terms of engagement, enjoyment, emotional expression, and whether they feel psychologically safe (RQ2).

Third, we look at what concrete opportunities and challenges appear when UX researchers take on a “game master” (GM) role during user studies (RQ3).

To answer these questions, we will use three main steps.

The first step is framework adaptation. Here we build a clear link between TTRPG mechanics and UX research activities. We systematically map core TTRPG ideas such as Character, Attributes, Challenge, Adventure, and GM to UX elements such as Persona, User Skills, Usability Task, User Journey, and UX Researcher. In this step, we also design the challenge mechanism: we set the difficulty level for key interactions in the app prototype, define how a Task Check works, and decide what kinds of “complications” happen in the story or task flow when a Challenge Check fails.

The second step is toolkit design. Based on the framework, we turn the ideas into concrete tools in Figma. We create a Player Attribute Sheet that defines the user’s “character” with 3–5 key attributes (for example, Patience, Domain Knowledge, Age, Gender) and a limited number of points to assign. We design an interactive Challenge Interface in Figma that simulates dice rolls and shows results. We also build scenario-based storyboards that group standard usability tasks into clear task flows. The storyboards show goals, steps, and where challenges happen, so the method is structured even when we do not use heavy narrative.

The third step is a user test. We run structured sessions where each participant uses the toolkit once. In each session, a UX researcher guides the participant through app-related tasks using the storyboards. The participant first creates or adjusts their Player Attribute Sheet to define their “user character.” Then they interact with the prototype and use the dice-based Challenge Interface at key points while completing the tasks in the storyboard. We record challenge results and feedback during the session.

For data collection, we focus on both numbers and stories. On the quantitative side, we record task success and failure based on Challenge Checks, time on task, how participants allocate attributes, how often complications are triggered, and post-session Likert-scale ratings of usability, clarity, engagement, enjoyment, and psychological safety. On the qualitative side, we collect participants’ comments on each toolkit component (attribute sheet, challenge interface, storyboards), including what they found useful or confusing and what they would improve.

In the analysis, we look at how participants understand and work with the toolkit, whether the TTRPG framing helps them express needs, frustrations, motivations, and emotions in more detail, how smoothly the toolkit fits into typical UX research tasks, and what difficulties researchers face when acting as GMs. The results will help us refine the toolkit and will be used to answer RQ1–RQ3, with a focus on insight quality, participant experience, and researcher facilitation challenges.

Expected Outcomes

By the end of the project, we expect to have three main kinds of outcomes.

First, we will have a TTRPG-inspired UX research framework. This will include a clear conceptual mapping between key TTRPG elements, such as characters, adventures, game masters, and challenges, and common UX research activities, including personas, scenarios, researcher roles, and usability tasks. Based on this mapping, we will also summarize practical guidelines that explain how small UX teams can plan, run, and facilitate TTRPG-style sessions in their everyday work.

Second, we will have a concrete set of gamified UX research tools built as Figma prototypes. These will take the form of reusable and customizable templates, including player attribute/persona character sheets, dice-and-challenge interfaces, and scenario-based storyboards that can be used to structure app evaluation sessions.

Third, we will produce empirical evidence and design implications in the form of a research report. This report will compare TTRPG-inspired methods with more conventional approaches, show examples of user needs that appeared only in the TTRPG condition or became clearer there, and reflect on participants' experiences and ethical considerations. It will also offer recommendations on when TTRPG-inspired approaches are especially helpful, and when simpler, traditional methods may be enough.

Timeline

The project is expected to take approximately 9 weeks. In Week 1, we will develop the TTRPG-inspired UX research framework. In Weeks 2–4, we will focus on designing the research tools and building the Figma prototypes. In Weeks 5–6, we will conduct user testing, including recruiting participants and running the full toolkit sessions. In Week 7, we will analyze the data collected from the user tests. Finally, in Weeks 8–9, we will synthesize the findings, draw conclusions, and complete the written report and paper. The schedule may be adjusted slightly in practice, but the overall sequence of activities will remain the same.

Overall, this project aims to make a first step toward turning TTRPG-based UX methods into something that can be used in everyday practice, helping to bridge the gap between the theoretical potential identified by Michael et al. and concrete, ready-to-use tools for user needs analysis.